

1)

```
plot(t,sum);
xlabel('time domain');
ylabel('f(t)');

x=1:6;
w=x.*pi;
v=2./x;%[More than one value then . (dot) is used, when single value then only / is used]
subplot(212);
stem(w,v);
xlabel('frequency domain');
ylabel('Amplitude');
```

2)

Q#01: Write a user defined MATLAB function to reconstruct $f(t)$ in the time interval of $[-4, 20]$ using the following Fourier series; also draw Discrete Frequency Spectrum

$$f(t) = 2.5 + \frac{10}{\pi} \left(\sin \frac{\pi t}{4} + \frac{1}{3} \sin \frac{3\pi t}{4} + \frac{1}{5} \sin \frac{5\pi t}{4} + \frac{1}{7} \sin \frac{7\pi t}{4} + \dots \right)$$

$$\underbrace{f(t)}_{\text{Complex wave}} = \underbrace{2.5}_{\text{DC value}} + \frac{10}{\pi} \sin \frac{\pi t}{4} + \frac{10}{3\pi} \sin \frac{3\pi t}{4} + \frac{10}{5\pi} \sin \frac{5\pi t}{4} + \frac{10}{7\pi} \sin \frac{7\pi t}{4} + \dots$$

$$\underbrace{f(t)}_{\text{Complex wave}} = \underbrace{2.5}_{\text{DC value}} - \underbrace{\frac{5}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} (\cos n\pi - 1)}_{\text{Amplitude}} \underbrace{\sin \frac{n\pi}{4}}_{\text{Frequency}} t \dots \dots \dots \text{(ii)}$$

Answer:

```
function [result] = zamal(n)
or
function zashim(n)
t = -4:0.001:20;
y = 2.5;
sum = y;
for(i = 1:2:n)
sum = sum + (10/pi)*(1/i)*sin(i*pi*t/4)
end
figure(1);
plot(t,sum);
x=1:6
w=x.*pi/4
v=(((-5/pi)*1./x).*(cos(x*pi))-1)
stem(w,v)
```

3)

Q#02: Write a user defined MATLAB function to reconstruct $f(t)$ in the time interval of $[-4, 16]$ using the following Fourier series and also draw Discrete Frequency Spectrum

$$f(t) = \pi - \sum_{n=1}^{500} \frac{2}{n} \sin n \pi t$$

Answer:

```
function [result]= four1(n)
```

```
t=-4:.001:16;
```

```
y=pi;
```

```
sum=y;
```

```
for(i=1:1:n)
```

```
sum = sum-((2/i)*sin(i*pi*t));
```

```
end
```

```
subplot(211);
```

```
plot(t,sum);
```

```
xlabel('time domain');
```

```
ylabel('f(t)');
```

```
x=1:6;
```

```
w=x.*pi;
```

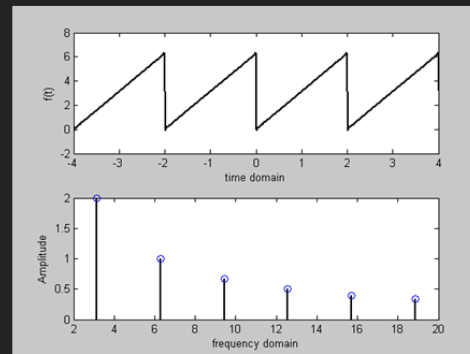
```
v=2./x; % [ More than one value then . (dot) is used, when single value then only / is used ]
```

```
subplot(212);
```

```
stem(w,v);
```

```
xlabel('frequency domain');
```

```
ylabel('Amplitude');
```



```

function zamal(n)
t=-4:.001:16;
y=pi;
sum=y;
for(i=1:1:n)
sum = sum-((2/i)*sin(i*pi*t));
end
subplot(211);
plot(t,sum);
xlabel('time domain');
ylabel('f(t)');

x=1:6;
w=x.*pi;
v=2./x;%[More than one value then . (dot) is used, when single value then only / is used]
subplot(212);
stem(w,v);
xlabel('frequency domain');
ylabel('Amplitude');

```

Q-09 (See Page # 67-72 from your lecture sheet)

$$\underbrace{f(t)}_{\text{Complex wave}} = \underbrace{2.5}_{\text{DC value}} + \underbrace{\left[-\frac{5}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} (\cos n\pi - 1) \sin \frac{n\pi t}{4} \right]}_{\text{AC value}}$$

$$\underbrace{f(t)}_{\text{Complex wave}} = \underbrace{2.5}_{\text{DC value}} + \left[-\frac{5}{\pi} \frac{1}{1} (\cos 1\pi - 1) \sin \frac{1\pi t}{4} - \frac{5}{\pi} \frac{1}{2} (\cos 2\pi - 1) \sin \frac{2\pi t}{4} - \frac{5}{\pi} \frac{1}{3} (\cos 3\pi - 1) \sin \frac{3\pi t}{4} - \frac{5}{\pi} \frac{1}{4} (\cos 4\pi - 1) \sin \frac{4\pi t}{4} + \dots \right]$$

$$\underbrace{f(t)}_{\text{Complex wave}} = \underbrace{2.5}_{\text{DC value}} + \left[-\frac{5}{\pi} (-1 - 1) \sin \frac{\pi t}{4} - \frac{5}{\pi} \frac{1}{2} (1 - 1) \sin \frac{2\pi t}{4} - \frac{5}{\pi} \frac{1}{3} (-1 - 1) \sin \frac{3\pi t}{4} - \frac{5}{\pi} \frac{1}{4} (1 - 1) \sin \frac{4\pi t}{4} + \dots \right]$$

$$\underbrace{f(t)}_{\text{Complex wave}} = \underbrace{2.5}_{\text{DC value}} + \frac{10}{\pi} \sin \frac{\pi t}{4} - \frac{5}{\pi} \frac{1}{2} (0) \sin \frac{2\pi t}{4} + \frac{10}{3\pi} \sin \frac{3\pi t}{4} - \frac{5}{\pi} \frac{1}{4} (0) \sin \frac{4\pi t}{4} + \dots$$

$$\underbrace{f(t)}_{\text{Complex wave}} = \underbrace{2.5}_{\text{DC value}} + \frac{10}{\pi} \sin \frac{\pi t}{4} + \frac{10}{3\pi} \sin \frac{3\pi t}{4} + \frac{10}{5\pi} \sin \frac{5\pi t}{4} + \frac{10}{7\pi} \sin \frac{7\pi t}{4} + \dots$$

$$f(t) = 2.5 + \frac{10}{\pi} \left(\sin \frac{\pi t}{4} + \frac{1}{3} \sin \frac{3\pi t}{4} + \frac{1}{5} \sin \frac{5\pi t}{4} + \frac{1}{7} \sin \frac{7\pi t}{4} + \dots \right)$$

```

t=-4:.001:20;
y=2.5;
z=(10/pi)*sin(pi*t/4);
p=(10/pi)*(1/3)*sin(3*pi*t/4);
q=(10/pi)*(1/5)*sin(5*pi*t/4);
r=(10/pi)*(1/7)*sin(7*pi*t/4);
s=(10/pi)*(1/9)*sin(9*pi*t/4);
u=(10/pi)*(1/11)*sin(11*pi*t/4);
v=(10/pi)*(1/11)*sin(11*pi*t/4);
x=y+z+p+q+r+s+u+v;
result=[t;z;p;q;r;s;u;v;x];
plot(t,x)

```

6)

Q-13:

$$x = \frac{1}{4} \cos 8t + \frac{1}{8} \sin 8t$$

```

A1=1/4;
B1=1/8;
t=0:0.001:1;
w=8;
y=A1*cos(w*t);
z=B1*sin(w*t);
x=y+z;
result=[t;y;z;x];
plot(t,y,t,z,t,x)

```

7)

$x[n] = 1, n = 0$ and $h[n] = 2, n = 0$
 $3, n = 1$ $1, n = 1$; n represents the time index

MATLAB

`x=[1 3]`

`h=[2 1]`

`y=conv(x,h)`

`y= [2 7 3]`

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8)

Q#11:

Write a code for the following function:

$4u(t) - 8u(t-1) + 4u(t-2)$

Answer:

`t = - 5:.01:10;`

`t0 = 0;`

`y = 4*stepfun(t,t0); % step function means 1 at  $t \geq 0$`

`t1=1;`

`z = -8*stepfun(t, t1);`

`t2=2;`

`x = 4*stepfun(t, t2);`

`w=y+z+x;`

`plot(t,w)`

Q#03: Write a MATLAB function to reconstruct $f(t)$ in the time interval of $[-4,16]$ using the following Fourier series and also draw Discrete Frequency Spectrum

$$f(t) = \frac{\pi^2}{3} + 4 \sum_{n=1}^{500} \frac{(-1)^n}{n^2} \cos n\pi t$$

```
function[result]=four(n)
t=-10:0.001:10;
y=(pi^2)/3;
sum=y;
for(i=1:1:n)
    sum=sum+(4*((-1)^i)/i^2)*cos(i*pi*t);
end
subplot(211);
plot(t,sum);
xlabel('time domain');
ylabel('f(t)');
x=1:10;
w=x.*pi;
v=(4*(-1).^x./x.*x);
subplot(212);
stem(w,v);
xlabel('frequency domain');
ylabel('amplitude');
```

**question :

Q#03: Write a MATLAB function to reconstruct $f(t)$ in the time interval of $[-4,16]$ using the following Fourier series and also draw Discrete Frequency Spectrum

$$f(t) = \frac{\pi^2}{3} + 4 \sum_{n=1}^{500} \frac{(-1)^n}{n^2} \cos n\pi t$$